

William Bengen's SAFEMAX

For sustainable spending from an investment portfolio, William Bengen's work is the next natural stop on our journey. William Bengen's seminal study in the October 1994 *Journal of Financial Planning*, "Determining Withdrawal Rates Using Historical Data," helped usher in the modern area of retirement withdrawal rate research by codifying the importance of sequence-of-returns risk. The problem he set up is simple: a newly retired couple plans to withdraw an inflation-adjusted amount from their savings at the end of every year for a thirty-year retirement period. What is the highest annual sustainable percentage of retirement date assets that can be withdrawn with inflation adjustments for a full thirty years?

To answer this question, Bengen obtained a copy of Ibbotson Associates' *Stocks, Bonds, Bills, and Inflation* yearbook, which provides monthly data for a variety of US asset classes and inflation since January 1926. He decided to investigate using the S&P 500 index to represent the stock market and intermediate-term government bonds to represent the bond market.

He constructed rolling thirty-year periods from this data (1926 through 1955, then 1927 through 1956, and so on), using a technique called "historical simulations." He calculated the maximum sustainable withdrawal rate for each rolling historical period. Such an approach helps illustrate the role of market volatility in a way that assuming a constant portfolio return does not. Though he did not create the following illustration, his spreadsheet calculations would provide something similar to what is seen in Exhibit 4.1.

Exhibit 4.1

Maximum Sustainable Withdrawal Rates

*For 50/50 Asset Allocation, 30-Year Retirement, Inflation Adjustments
Using SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government
Bonds*